

Answer all questions.

NEITHER MATHEMATICAL TABLES NOR SLIDE RULES
NOR CALCULATORS MAY BE USED IN THIS PAPER.

- 1 (a) Express 4 560 in standard form.

Answer (a)

[1]

- (b) It is given that $2 \times 10^3 + 1 \times 10^1 + 3 \times 10^x + 5 \times 10^y = 2\,013,05$
where x and y are integers.
Find the value of x and the value of y .

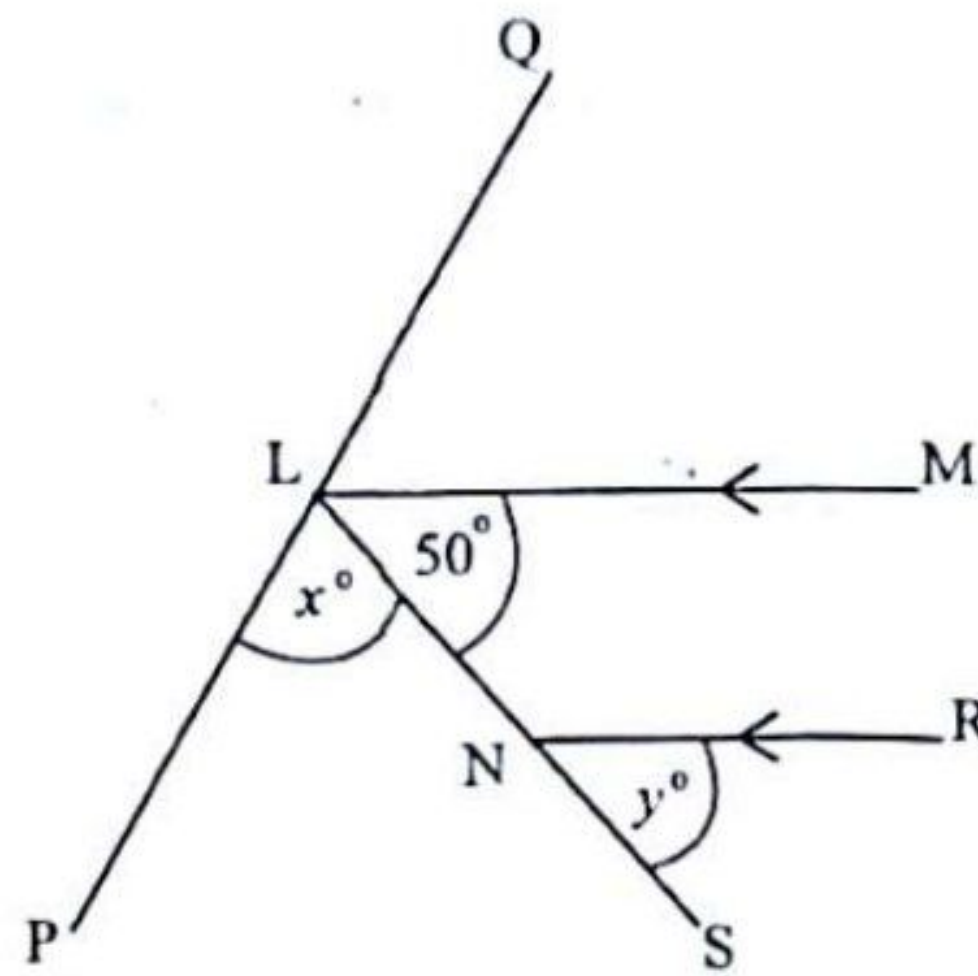
Answer (b)

[2]

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In the diagram, PLQ is a straight line. Line LM is parallel to line NR. The line LM bisects $\angle QLN$. $\angle NLM = 50^\circ$. $\angle PLN = x^\circ$. $\angle RNS = y^\circ$

Find the angle marked

(a) x ,

Answer (a)

[2]

(b) y .

Answer (b)

[1]

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[Turn over

- 3 The temperature reading in $^{\circ}\text{C}$, at noon, over a period of 6 days were 30, 29, x , 31, 30 and 32.

The mean temperature for the 6 days was $30,5^{\circ}\text{C}$.

Calculate the value of x .

[3]

- 4 (a) Solve the inequality $x - 8 > 4x + 1$.

Answer (a)

[2]

- (b) Represent the solution in (a) on a number line.

Answer (a)

[1]

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5 Evaluate

(a) $0,2 \times 0,35$,

Answer (a)

[1]

(b) $1\frac{2}{5} \div 1\frac{2}{3}$,

Answer (b)

[2]

(c) $\sqrt{0,0081}$.

Answer (c)

[1]

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[Turn over]

- 6 The temperature readings at 09 00 and 16 00 were -3°C and 18°C respectively. It is assumed that the temperature was rising at a steady rate.

Find the

- (a) temperature range,

Answer (a)

[1]

- (b) rate of temperature change per hour,

Answer (b)

[1]

- (c) temperature at 13 00.

Answer (c)

[2]

- 7 (a) Round off 148,32 to the nearest ten.

Answer (a)

[1]

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- (b) Estimate $179,35 \times 0,0496$, by expressing each of 179,35 and 0,0496, to one significant figure.

Answer (b)

[3]

8

A solid cuboid measures 5cm by 2cm by x cm.

- (a) Form an expression, in terms of x , for the total surface area of the cuboid.

Answer (a)

[2]

- (b) (i) The total surface area of the cuboid is 62cm^2 .
Form an equation in x .

Answer (b)(i)

[1]

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[Turn over]

- (ii) Solve the equation in (b)(i) to find the value of x .

Answer (b)(ii)

[1]

- 9 (a) A solid object has a mass of 300kg and a volume of 2m^3 .
Calculate the density of the object.

Answer (a)

[2]

- (b) A solid object has a mass of 300kg and a volume of 2m^3 .
Calculate the volume of a similar object given that its mass is 975kg.

Answer (b)

[2]



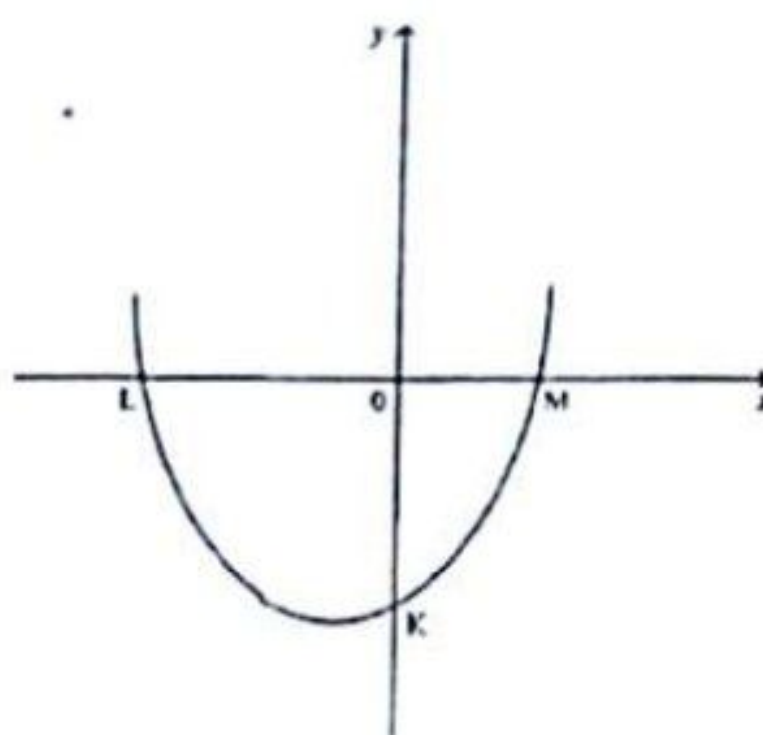
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The diagram is a sketch of $y = x^2 + x - 12$.

The graph cuts the y - axis at $(0 ; K)$

It crosses the x - axis at L and M.

- (a) Write down the value of K.

Answer (a)

[1]

- (b) Find the length of LM.

Answer (b)

[3]

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[Turn over]

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11 y is directly proportional to the square root of x .

- (a) Given that $x = 8$ when $y = 16$,
find the formula connecting x and y ,

Answer (a)

[2]

- (b) Find the value of x when $y = 12$.

Answer (b)

[2]

12 (a) Evaluate

(i) $\left(\frac{1}{3}\right)^{-2}$,

Answer (a)(i)

[1]

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(ii) $4^2 + 4^0$.

Answer (a)(ii)

[1]

(b) Solve the equation $5^{a-1} = 125$.

Answer (b)

[2]

13 (a) Factorise completely $5m^2 - 20$.

Answer (a)

[2]

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[Turn over]



(b) Simplify $\frac{3x + 3y}{x^2 - y^2}$.

Answer (b)

[2]

- 14 The table below shows the number of children living in houses at the staff quarters at a certain school.

Number of children in a house	0	1	2	3	4	5
Number of houses	2	3	4	5	1	0

Find the

- (a) mode,

Answer (a)

[1]

- (b) median,

Answer (b)

[1]

- (c) mean.

Answer (c)

[2]



(b) Simplify $\frac{3x + 3y}{x^2 - y^2}$.

Answer (b)

[2]

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Answer (b)

[1]

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Answer (c)

[2]

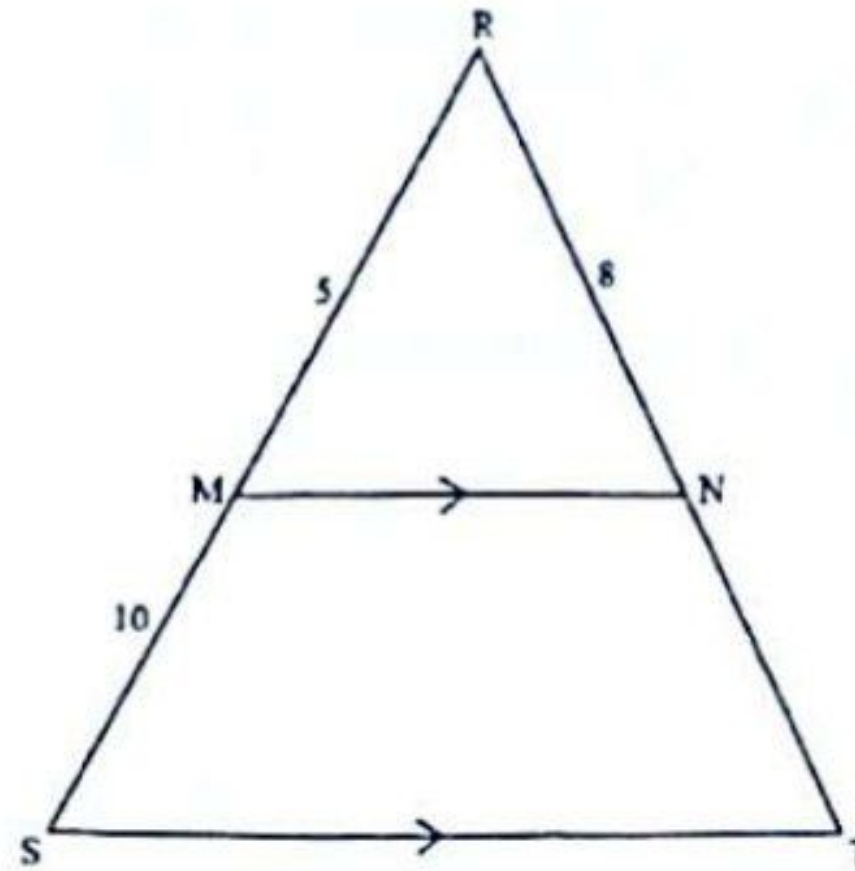
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13

15



In the diagram is triangle RST.

MN is parallel to ST.

$RM = 5 \text{ cm}$, $RN = 8 \text{ cm}$ and $MS = 10 \text{ cm}$

- (a) Name the triangle similar to triangle RST.

Answer (a)

[1]

- (b) Find the length of NT.

Answer (b)

[3]

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[Turn over]

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16 Given that $\vec{AB} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ and $\vec{BC} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$,

(a) express $\vec{AC}$ as a column vector.

Answer (a)

[2]

(b) find $|\vec{AB}|$.

Answer (b)

[2]

17 (a) Convert 110101_2 to a number in base five.

Answer (a)

[2]

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- (b) Arrange the following numbers in ascending order
 11010_2 , 312_5 and 13_{10}

Answer (b)

[2]

- (c) Evaluate $312_5 - 143_5$. Give the answer in base 5.

Answer (c)

[1]

- 18 The scale on a map is given as 1 : 150 000.

Find the

- (a) distance on the ground represented by 1 cm on the map,
Give the answer in km.

Answer (a)

[1]

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[Turn over]



- (b) distance on the map representing a distance on the ground of 7,5 km.
Give the answer in cm.

Answer (b)

[2]

- (c) actual area represented by an area of 4 cm^2 on the map.
Give the answer km^2 .

Answer (c)

[2]

19 It is given that $A = h(4m + h)$.

- (a) Express m in terms of h and A ,

Answer (a)

[3]

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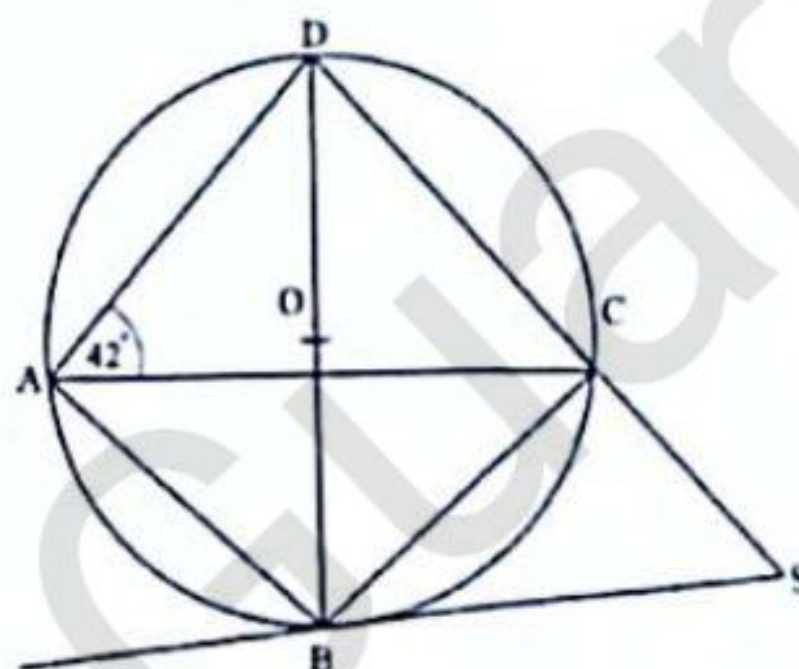
17

- (b) Given that $A = 20$ and $h = 2$,
calculate the value of m .

Answer (b)

[2]

20



The diagram shows points A, B, C and D on the circumference of a circle centre O.
The line BS is a tangent to the circle at B.
DC produced meet the tangent from B at S
 $\angle DAC = 42^\circ$.

Find

- (a) $\angle OBC$,

Answer (a)

[1]

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[Turn over]



(b) $\hat{CAB}$,

Answer (b)

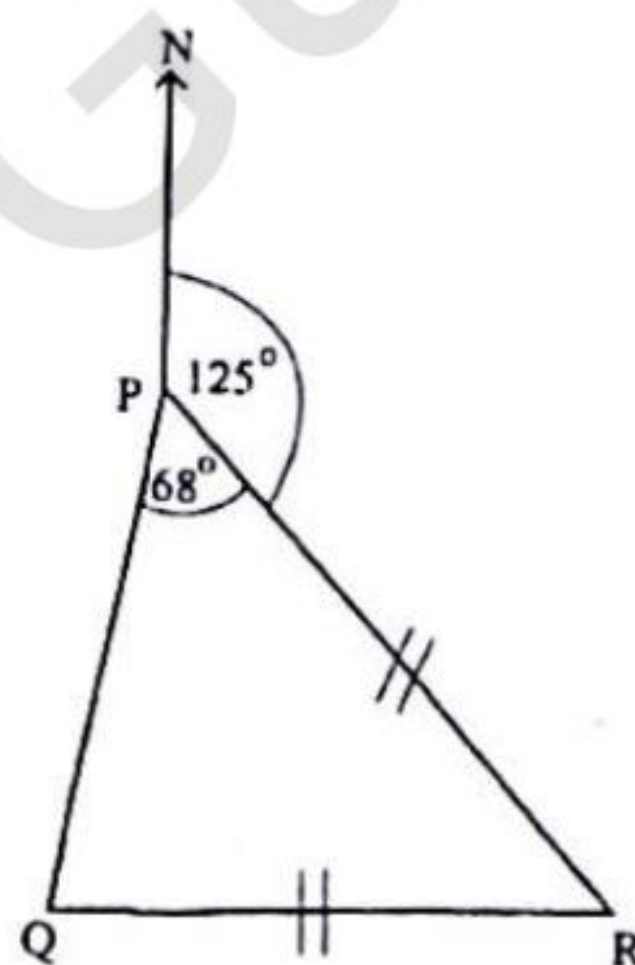
[2]

(c) $\hat{DSB}$,

Answer (c)

[2]

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The diagram shows 3 towns P, Q and R on level ground.

R is equidistant from P and Q.

The bearing of R from P is 125° .

$\angle PQR = 68^\circ$.

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Find the

(a) angle $\widehat{PRQ}$,

Answer (a)

[2]

(b) bearing of P from R,

Answer (b)

[1]

(c) bearing of P from Q.

Answer (c)

[2]

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[Turn over



20

22 Given that $4P - 2 \begin{pmatrix} 3 & 1,5 \\ -6 & 0 \end{pmatrix} = P$,

find

- (a) matrix P in the form $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where a, b, c and d are numbers,

Answer (a)

[3]

- (b) the inverse of matrix P .

Answer (b)

[2]

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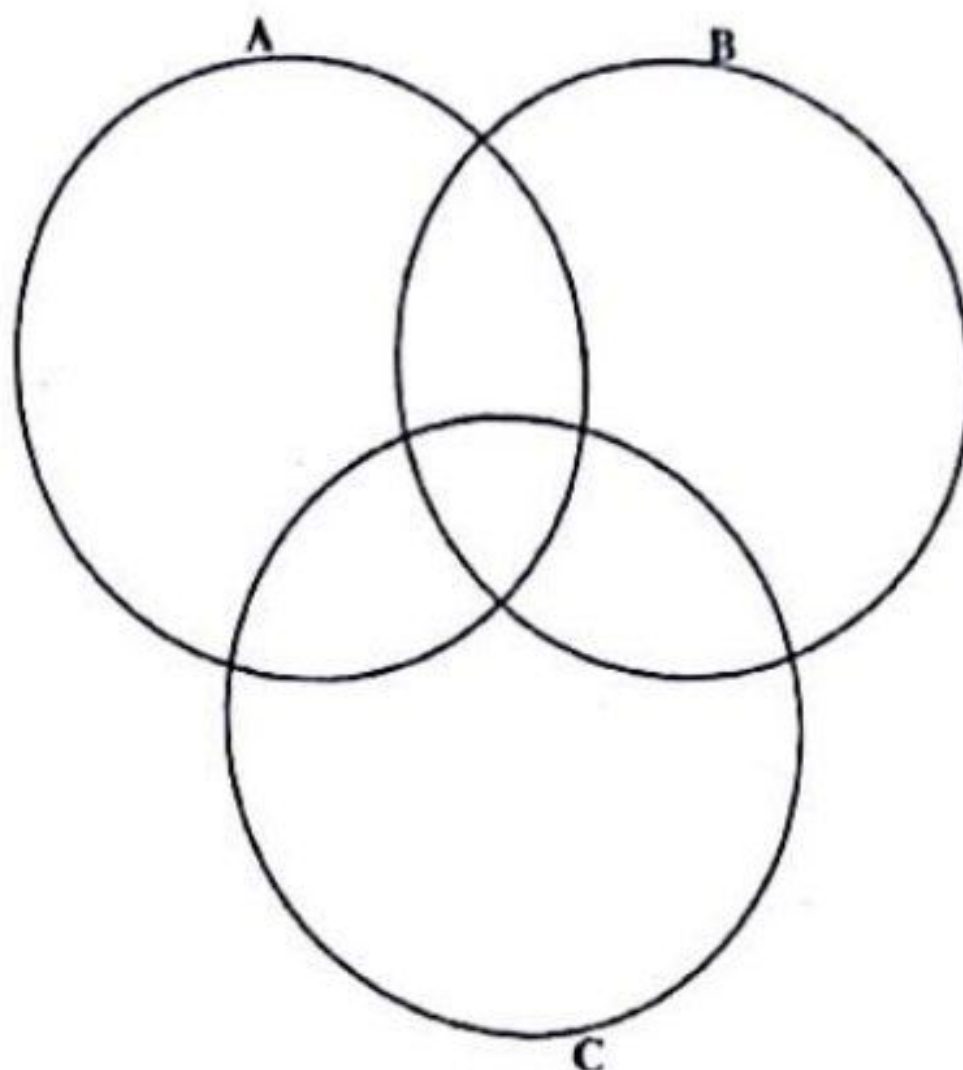


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23

(a)



On the Venn Diagram above shade the set $(A \cap B) \cup C$.

Answer(a)

[2]

(b)

In a class of 45 children,
30 do Agriculture.

25 do Computers.

7 do neither Agriculture nor Computers.

Find the number of children who do Agriculture only.

Answer (a)

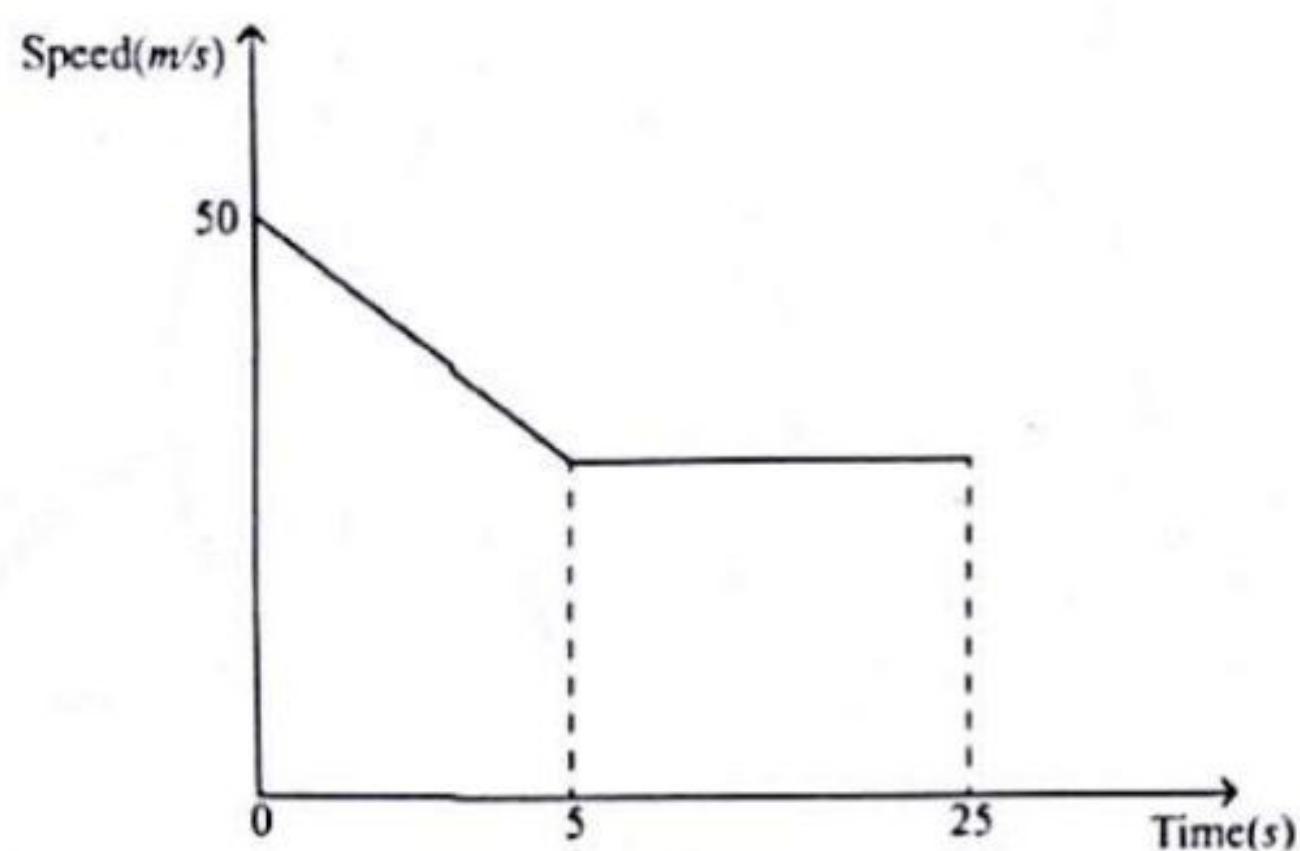
[3]

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[Turn over]



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The diagram is a speed - time graph of a moving object.

The starting speed is 50m/s.

It decelerates at 4m/s for the first 5 seconds

It maintains constant speed for the next 20 seconds.

- (a) Calculate the speed of the object when time is 10 seconds.

Answer (a)

[3]

- (b) Calculate the distance travelled during the 25 seconds.

Answer (a)

[2]

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